SunCell Backup | spec sheet

Tesla Lab Product Design Engine - numbers computed in Python, not guessed

Solar panels make hydrogen by day; a fuel cell keeps a rural clinic's fridge running through long outages.

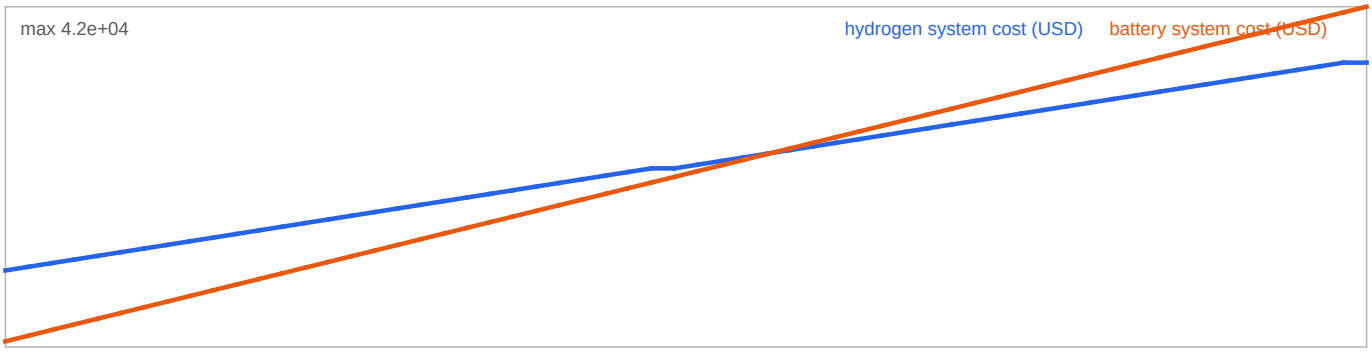
Derived from the claim that WORKS: Hydrogen made from water by electrolysis can store energy and power a vehicle, if the electricity comes from outside.



Key numbers

- 48 h of 150 W backup needs 0.43 kg of hydrogen in 4 x 50 L tanks at 30 bar
- Round trip 30%: 23.8 kWh of solar in, 7.2 kWh out
- Refill in 7 days with 0.9 kWp of extra solar and 3.9 L of water
- Honest comparison: a 8 kWh battery costs about $2,800; hydrogen only wins for outages over 18 days
- Prototype parts ~ $11,514 (estimate)

hydrogen system cost (USD) vs backup duration (h)



Requirements

Average critical load	**150 W**	req
Peak load	**400 W**	req
Backup duration	**48 h**	req

Refill the tank in	**7 days**	req

Energy

Energy delivered per outage	**7.2 kWh**	calc
Fuel cell efficiency (LHV)	**50 %**	assumed
Hydrogen needed LHV 33.33 kWh/kg	**0.432 kg**	calc

Storage

Storage pressure electrolyser outlet pressure, no compressor	**30 bar**	req
H2 density at that pressure (25 degC)	**2.395 kg/m³**	calc
Tank volume needed 93% usable	**193 L**	calc
50 L cylinders	**4**	calc
Water consumed per refill 8.94 kg water per kg H2	**3.9 L**	calc

Charging

Electrolyser energy per refill 55.0 kWh/kg (assumed, small PEM)	**23.8 kWh**	calc
Round-trip efficiency	**30.3 %**	calc
Electrolyser size	**0.75 kW**	calc
Extra solar PV needed 4.5 peak-sun-hours/day (assumed, typical Malaysia), performance ratio 0.8	**0.94 kWp**	calc

Output

Fuel cell rating	**400 W**	calc

Battery comparison

LFP battery for the same outage	**8 kWh**	calc
Battery cost / H2 system cost	**$2,800 / $10,800**	est.
H2 becomes cheaper for outages longer than	**17.5 days**	est.

Bill of materials (estimates)

PEM electrolyser ~0.8 kW, 30 bar outlet	**1 x $5,500.00**	$5,500
PEM fuel cell 400 W with DC-DC	**1 x $2,500.00**	$2,500
50 L steel cylinder, 30 bar rated	**4 x $450.00**	$1,800
Solar PV, 0.9 kWp extra	**0.94 x $600.00**	$564
Water purifier / deioniser	**1 x $250.00**	$250
Ventilated outdoor cabinet, H2 sensor, valves, controls	**1 x $900.00**	$900

Total (estimate): $11,514

Rough estimates from typical 2025-26 distributor / retail list prices for one-off prototype quantities, not quotes. Verify before you buy.

Honest limits

Only about a third of the solar energy comes back (the math above). Batteries are better for daily cycling; hydrogen wins only for long, rare outages. It never makes energy from water.

Caveats

- Never makes energy from water: about 70% of the solar energy is lost in the loop.
- Component efficiencies and prices are assumptions for small (sub-kW) systems.
- Hydrogen storage needs ventilation, leak sensing and local code approval.

Render prompt (generated from these numbers)

Photoreal product render, 3/4 view, tropical late-afternoon light, palm trees softly blurred, no text. A white and teal ventilated outdoor cabinet about 1.6 m tall beside a small rural clinic; through a side window 4 slim grey 50-litre steel hydrogen cylinders stand upright. A compact electrolyser and a 400 W fuel cell module sit on shelves above; a small water tank on the side. 2 solar panels on the clinic roof, a status panel glowing green.

References

- US DOE Hydrogen Program, H2 properties (LHV 33.3 kWh/kg) https://www.energy.gov/eere/fuelcells/hydrogen-storage
- NIST Chemistry WebBook, hydrogen fluid properties https://webbook.nist.gov/chemistry/fluid/